COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE 010 Data Structures and Algorithms | **SECTION:** | CPE 21S2 |
| **DATE PERFORMED:** | July 10, 2026 | **DATE SUBMITTED:** | July 15, 2026 |
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**ACTIVITY NO. 1.1** **Hands on Activity: Basic C++ Programming**

## 1. PROCEDURE OUTPUT

ILO A:

ILO A.cpp

```cpp
#include<iostream>
using namespace std;

void displaySum(int a, int b);
bool isGreater( int A, int B);
bool logicalOperations(bool x, bool y);

int main() {
    int num1, num2;
    bool val1, val2;

    cout <<"Enter First Number:";
    cin >> num1;
    cout <<"Enter Second Number:";
    cin >> num2;

    displaySum(num1, num2);

    if (isGreater(num1, num2))
        cout << num1 << " is greater than " << num2 << endl;
    else
        cout << num1 << " is not greater than " << num2 << endl;

    cout << "Enter first boolean value (0 or 1) :";
```

```
28      cin >> val2;
29
30      if (logicalOperations(val1, val2))
31          cout << "Logical operations completed succesfully" << endl;
32
33
34      return 0;
35  }
36
37  void displaySum(int a, int b) {
38      cout << "\nSum = " << a + b << endl;
39
40  }
41
42  bool isGreater(int A, int B) {
43      return (A > B);
44  }
45
46  bool logicalOperations(bool x, bool y) {
47      cout << "\nLogical AND = " << (x && y) << endl;
48      cout << "Logical OR = " << (x || y) << endl;
49      cout << "Logical NOT of first value = " << (!x) << endl;
50      cout << "Logical NOT of second value = " << (!y) << endl;
51
52      return true;
53  }
54
55
```

Output:

```
Enter First Number: 67
Enter Second Number:99

Sum = 166
67 is not greater than 99
Enter first boolean value (0 or 1) :1
Enter second boolean value (0 or 1) :0

Logical AND = 0
Logical OR = 1
Logical NOT of first value = 0
Logical NOT of second value = 1
Logical operations completed succesfully
```

Code explanation:
In the code that I created, I used #inlucde <iostream> and using namespace std; to allow input and output in the

program using also bool and void commands in order to display the sum of the numbers and for bool for the logical numbers if its greater than the number or if its not a greater number and also used another bool for the logical operations.

| Sections | Answer |
|---|---|
| Header File Declaration Section | #include <iostream> using namespace std; |
| Global Declaration Section | No global variables declared |
| Class Declaration and Method Definition Section | void displaySum(int a, int b); bool isGreater(int A, int B); bool logicalOperations(bool x, bool y); |
| Main Function | cout << "Enter first boolean value (0 or 1) :";<br>cin >> val1;<br><br>cout << "Enter second boolean value (0 or 1) :";<br>cin >> val2;<br><br>if (logicalOperations(val1, val2))<br>cout << "Logical operations completed succesfully" << endl;<br><br>return 0;<br>}<br><br>void displaySum(int a, int b) {<br>cout << "\nSum = " << a + b << endl;<br><br>}<br><br>bool isGreater(int A, int B) {<br>return (A > B);<br>}<br><br>bool logicalOperations(bool x, bool y) {<br>cout << "\nLogical AND = " << (x && y) << endl;<br>cout << "Logical OR = " << (x \|\| y) << endl;<br>cout << "Logical NOT of first value = " << (!x) << endl;<br>cout << "Logical NOT of second value = " << (!y) << endl;<br><br>return true;<br>} |
| Method Definition | displaySum(), isGreater(), and logicalOperations() |

ILO B:

Code:

ILO B.cpp > ...

```cpp
#include<iostream>
class Triangle{
private:
double totalAngle, angleA, angleB, angleC;

public:
Triangle(double A, double B, double C);
void setAngles(double A, double B, double C);
const bool validateTriangle();
};
Triangle::Triangle(double A, double B, double C) {
angleA = A;
angleB= B;
angleC = C;
totalAngle = A+B+C;
}
void Triangle::setAngles(double A, double B, double C) {
angleA = A;
angleB = B;
angleC = C;
totalAngle = A+B+C;
}
const bool Triangle::validateTriangle() {
return (totalAngle <= 180);
}

int main(){
Triangle set1(40, 30, 110);
if(set1.validateTriangle()){
std::cout << "The shape is a valid triangle.\n";
} else {
std::cout << "The shape is NOT a valid triangle.\n";
}
return 0;
}
```

Output:

```
The shape is a valid triangle.
```

Code explanation:
In this code after I followed the instructions based on the activity of following the code using triangle as a class and putting a private command for the three angles: angles a, b and c then used the void command to set the angles and bool to validate if the shape is a valid triangle which as we saw in the output proves that the code works.

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C
Problem 1

ILO C 1.cpp > main()

```cpp
#include<iostream>
using namespace std;

void swapNumbers(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}

int main() {
    int num1, num2;

    cout << "Enter first number:";
    cin >> num1;

    cout << "Enter second number:";
    cin >> num2;

    cout << "\nBefore swapping:\n";
    cout << "First Number = " << num1 << endl;
    cout << "Second Number = " << num2 << endl;

    swapNumbers(num1, num2);

    cout << "\nAfter swapping:\n";
    cout << "First Number = " << num1 << endl;
    cout << "Second Number = " << num2 << endl;

    return 0;
}
```

```
Enter first number:55
Enter second number:98

Before swapping:
First Number = 55
Second Number = 98

After swapping:
First Number = 98
Second Number = 55
```

Code explanation:
In this code that I created, I used using namespace std; and then used void swapNumbers in order to swap the numbers of the first and second integer next is I used a lot of cout and cin to display and to enter the first number and second number then used the swapNumbers of the first number and the second number and observed that the code is working properly that the first number and the second number swapped in the after swapping section.

Problem 2:

ILO C 2.cpp > main()

```cpp
#include<iostream>
using namespace std;

double kelvinToFahrenheit(double kelvin) {
    return (kelvin - 273.15) * 9/5 + 32;
}

int main() {
    double kelvin;

    cout << "Enter Temperature in Kelvin: ";
    cin >> kelvin;

    double fahrenheit = kelvinToFahrenheit(kelvin);
    cout << kelvin << " Kelvin is equal to " << fahrenheit << " Fahrenheit." << endl;

    return 0;
}
```

Output:

```
Enter Temperature in Kelvin: 99
99 Kelvin is equal to -281.47 Fahrenheit.
```

Code explanation:
In this code that I created, I used namespace again and this time I used the double command to convert kelvin to Fahrenheit and used the return command to convert kelvin which is kelvin – 273.15 * 9/5 + 32 the used the double command on the main code then used cout in order to enter the temperature in kelvin before displaying the final conversion into Fahrenheit.

Problem 3:

Code:

ILO C 3.cpp > main()

```cpp
#include<iostream>
#include<cmath>
using namespace std;

double calculateDistance(double x1, double y1, double x2, double y2) {
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}

int main() {
    double x1, y1, x2, y2;

    cout << "Enter coordinates of first point (x1 y1): ";
    cin >> x1 >> y1;

    cout << "Enter coordinates of second point (x2 y2): ";
    cin >> x2 >> y2;

    double distance = calculateDistance(x1, y1, x2, y2);
    cout << "The distance between the two points is: " << distance << endl;
```

```cpp
    return 0;
}
```

Output:

```
Enter coordinates of first point (x1 y1): 9 3
Enter coordinates of second point (x2 y2): 6 7
The distance between the two points is: 5
```

Code explanation:
In this code that I created, I used this time is the cmath command and also used the double command in order to calculate the distance of x1, y1 and x2, y2 and before it shows the results the points must be entered manually in order to get the final result which is shown in the output.

Problem 4:

Code:

ILO C 4.cpp > Triangle(double, double, double, double, double, double)

```cpp
#include<iostream>
#include<cmath>
using namespace std;

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;
    double sideA, sideB, sideC;

public:
    Triangle(double A, double B, double C, double a, double b, double c);
    void setValues(double A, double B, double C, double a, double b, double c);
    bool validateTriangle();
    double getPerimeter();
    double getArea();
    void triangleType();

};
```

```cpp
Triangle::Triangle(double A, double B, double C, double a, double b, double c) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;

    sideA = a;
    sideB = b;
    sideC = c;

}

void Triangle::setValues(double A, double B, double C, double a, double b, double c) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
```

```
37
38      sideA = a;
39      sideB = b;
40      sideC = c;
41  }
42
43  bool Triangle::validateTriangle() {
44      return (totalAngle == 180);
45  }
46
47  double Triangle::getPerimeter() {
48      return sideA + sideB + sideC;
49  }
50  void Triangle::triangleType() {
51      if (angleA < 90 && angleB < 90 && angleC < 90) {
52          cout << "The triangle is an acute triangle." << endl;
53      } else if (angleA == 90 || angleB == 90 || angleC == 90) {
51      if (angleA < 90 && angleB < 90 && angleC < 90) {
53      } else if (angleA == 90 || angleB == 90 || angleC == 90) {
54          cout << "The triangle is a right triangle." << endl;
55      } else {
56          cout << "The triangle is an obtuse triangle." << endl;
57      }
58  }
59
60  int main() {
61      Triangle set1(40, 30, 110, 8, 7, 5);
62
63      if (set1.validateTriangle()) {
64          cout << "The shape is a valid triangle." << endl;
65          cout << "Perimeter of the triangle: " << set1.getPerimeter() << endl;
66          set1.triangleType();
67      } else {
68          cout << "The shape is NOT a valid triangle." << endl;
69      }
```

```
58  }
59
60  int main() {
61      Triangle set1(40, 30, 110, 8, 7, 5);
62
63      if (set1.validateTriangle()) {
64          cout << "The shape is a valid triangle." << endl;
65          cout << "Perimeter of the triangle: " << set1.getPerimeter() << endl;
66          set1.triangleType();
67      } else {
68          cout << "The shape is NOT a valid triangle." << endl;
69      }
70
71      return 0;
72  }
```

Output:

```
The shape is a valid triangle.
Perimeter of the triangle: 20
The triangle is an obtuse triangle.
```

Code explanation:
Based on the ILO C problem 4's instructions, I used the ILO B's code but added a few more lines of codes in the code of ILO B the contents that includes in the earlier problems from bool to validate, double to get the parameter of the triangle, and void for its triangle type and this time I added some conditions such as the if else, if the triangle is valid it will display that it wile valid and shows the parameter and if its acute, obtuse and right triangle if not it will display a "NOT a valid triangle"

## 3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

**Tools Used:** **ChatGPT**

*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

**Purpose of Use:** **To receive explanations of the required C++ code, programing concepts and proper syntax for each task in the activity**

*Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.*

**Extent of Use:** **I used AI to explain the purpose of the code and refresh my understanding of C++ basic programming. The explanations served as a reference while I wrote and modified the programs based on the activity requirements. I reviewed, tested and finalized the code myself.**

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Conclusion:

In this activity, I learned the basic concepts of C++ programming, including functions, classes, logical operators, and problem-solving through coding. The procedures and supplementary activities helped me improve my understanding of writing, testing, and explaining C++ programs. Overall, I think I did well in completing the tasks, but I still need to improve my coding efficiency, debugging skills, and understanding of more advanced C++ concepts.